PicoMagnetometer RPi software User manual

Pico nT

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Python code for the UKRAA PicoMagnetometer

Set of Python, gnuplot and shell scripts to run on an RPi4/5 to record, process and present data from the UKRAA PicoMagnetometer.

This software was written to suit a specific set-up; feel free to use as you see fit.

Instructions for initial setting up of a Raspberry Pi4/5 are included in the **docs** folder

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Using the code

The code assumes that your UKRAA PicoMagnetometer is the only device using USB that is connected to the RPi4/5 via supplied USB cable and that it is **/dev/ttyACM0** - you can check this by using **ls /dev/tty*** in a terminal window on the RPi4/5 and reviewing the response.

The code assumes username is **pi**.

If **pi** is not the username, then you will need to change all occurrences of '**/home/pi**' to '**/home/*username***' in the python, gnuplot and shell scripts, where ***username*** is the username you have selected for your RPi4/5.

The code assumes one magnetometer connected to the RPi4/5 USB and that it will be connected via **/dev/ttyACM0.**

If there are other devices connected to the RPi and your magnetometer is not **/dev/ttyACM0**, then you will need to change **/dev/ttyACM0** to **/dev/*ttyACMx*** in the **GetDataRawACM0.py** python script, where ***ttyACMx*** is the tty address of you connected magnetometer.

GetDataRawACM0.py is run as a service.

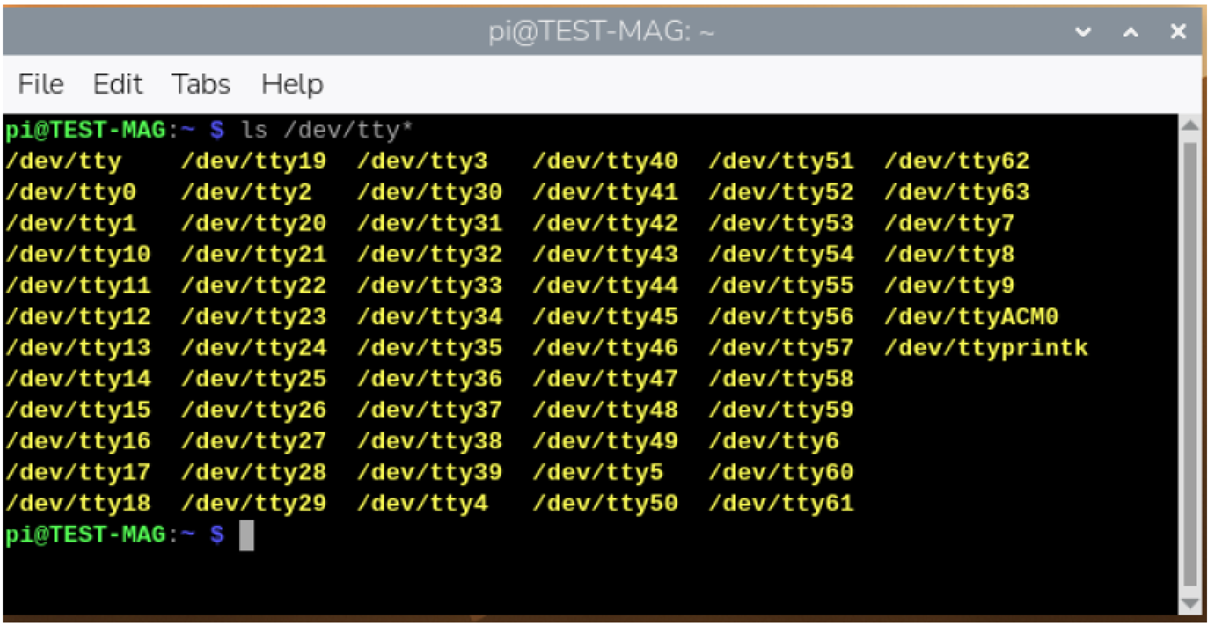
Other scripts (Python and gnuplot) are run from **cron** using shell scripts.

Where is my PicoMagnetometer?

Plug your PicoMagnetometer into any of the RPi USB ports - I normally use one of the blue ports (USB3).

1. Open a terminal window and type the following command and press enter

   ls /dev/tty*



2. You are looking for **/dev/ttyACM0** - this is on the right hand side of the screen shot above.

3. This is the USB address for your attached PicoMagnetometer - if you have more than one magnetometer attached you may see **/dev/ttyACM1** etc.

4. If you do not see **/dev/ttyACM0**, then unplug and plug the PicoMagnetometer back in and try again.

5. As long as you see **/dev/ttyACM0** then you do not have to make any changes to the python scripts, because they are looking for **ACM0**.

Getting the software onto your RPi

1. Log into your Raspberry Pi4/5 using VNC.

2. Open a terminal window, type the following command and press enter

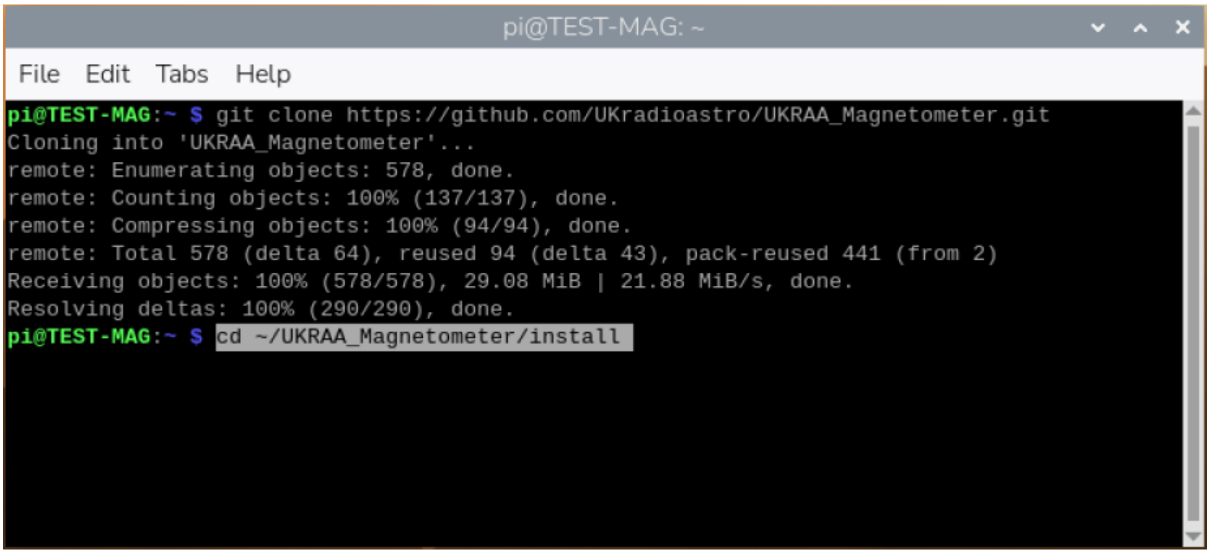
   git clone https://github.com/UKradioastro/UKRAA_Magnetometer.git

This will download all of the code to the directory **UKRAA_Magnetometer** inside **/home/pi**

Installing the software onto your RPi

1. Open a terminal window and type the following command and press **enter**

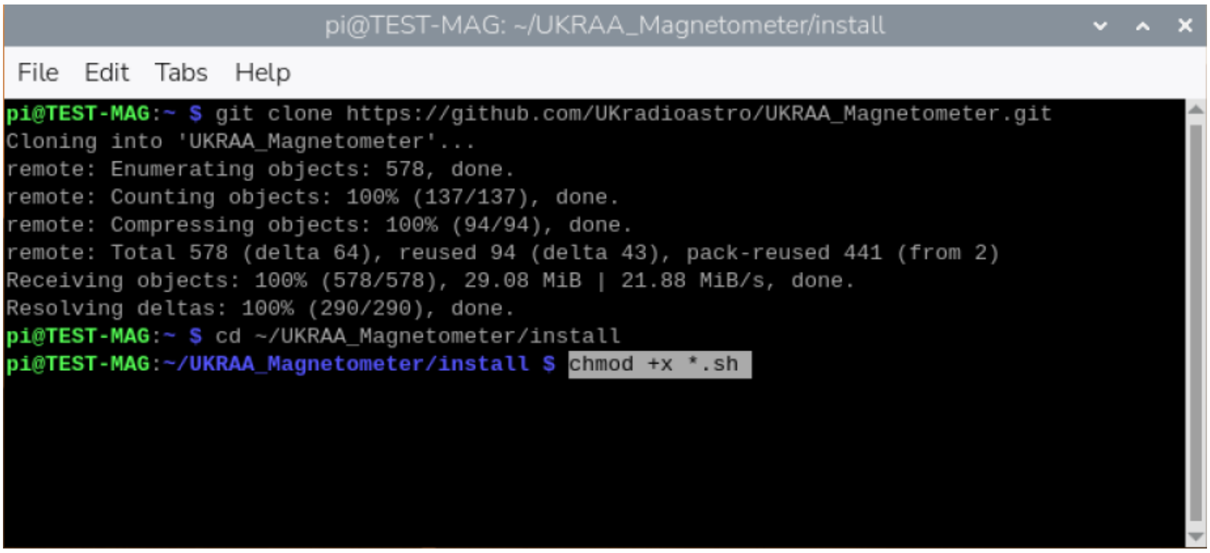
   cd ~/UKRAA_Magnetometer/install



This will take you to the **install** directory inside **/home/pi/UKRAA_Magnetometer**

2. Type the following command and press **enter**

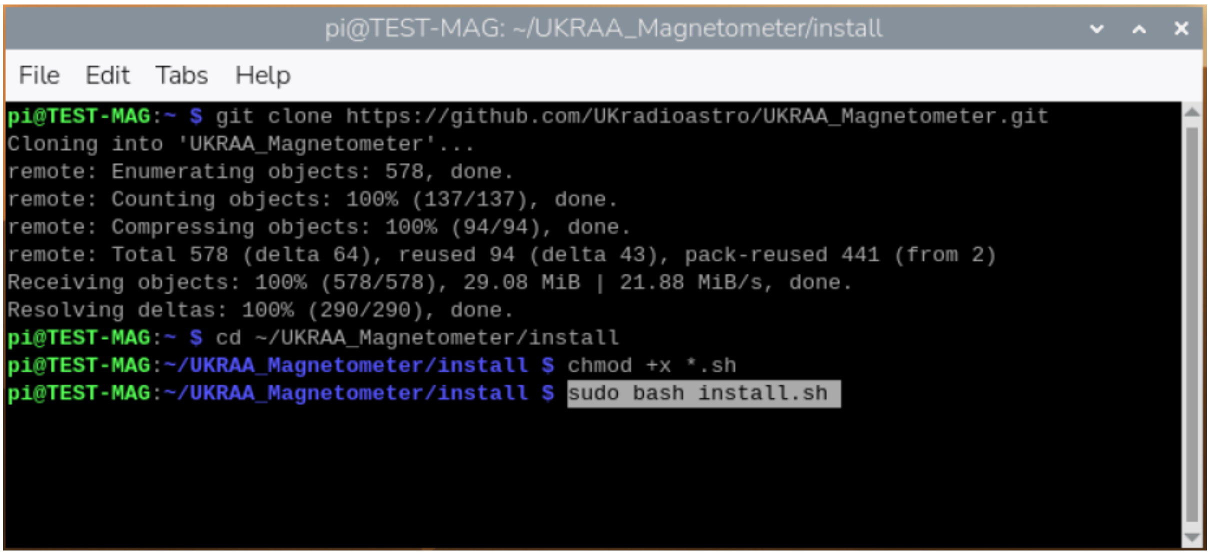
   chmod +x *.sh



This will make the **install.sh** script executable.

3. Type the following command and press **enter**

   sudo bash install.sh



This will run the install script.

There may be occasions during the running of the install script that require you to make a keyboard entry.

When asked **Do you want to continue? [Y/n]** - type **Y** or **y** and press **enter**

That's it!

The code is now set up to run automatically; it will record the data from the PicoMagnetometer, process the data, plot the data and post the plots to your intranet web page.

There are other functions that are customisable by the user, such as **email alerts** and **FTP** to users' external website, that are covered in the **User Manual**.

Optional: run an install-time heartbeat smoke check (disabled by default):

sudo MAGNETOMETER_INSTALL_SMOKE_HEARTBEAT=1 bash install.sh

This runs **--test-heartbeat** once during install and prints a clear **HEARTBEAT_SMOKE_CHECK: PASS** or **HEARTBEAT_SMOKE_CHECK: FAIL** summary.

What does the code do?

The code records data from the UKRAA PicoMagnetometer via serial over the supplied USB cable and stores the data to the raw data folder.

The raw data will be processed daily, via CRON, to get magnetic field values per minute in the X (N->S), Y (E->W) and Z (U->D) directions from your previous day's data. The maximum hourly variation of the magnetic field values in the X (N->S) and Y (E->W) directions will also be produced.

A number of plots will be created:

- X, Y and Z (North/South, East/West and up/down)
- Maximum hourly variation in X and Y directions
- H, D and Z (Local horizontal plane, declination angle and up/down)
- B and I (Total strength of Earth's magnetic field and angle of Earth's magnetic field)

Note: H, D & Z plots together with B & I plots are disabled by default, these plots can be generated by changing the configuration files - see **User Manual: Additional graphs** for more details.

The raw data will also be processed on a continuous 5 minute basis, again via CRON, to generate a rolling 24 hour plot of X, Y and Z magnetic fields and % change of magnetic field for combined X and Y directions. The latter used to predict the potential of visible Aurora activity. Should a threshold level be reached for % change of magnetic field, then the user will receive an email alert of such - see **Optional Rolling alert emails** section for more details.

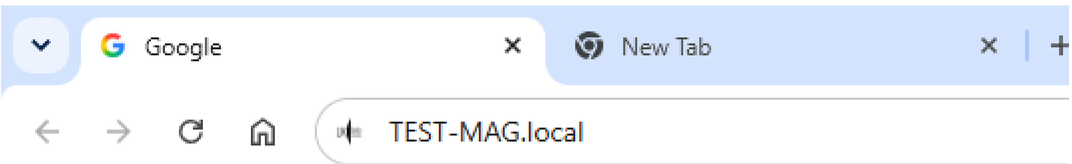
A simple web server and web page is set up on your RPi so that you can view your PicoMagnetometer results on your desktop PC and/or smart phone when connected to your home network.

To access the PicoMagnetomer webpage from your desktop PC or your smart phone…

1. Open your preferred web application (Safari, Chrome, Firefox, etc.).

2. In the search bar type the following and press enter

   http://test-mag.local



This will take you to the web page for your PicoMagnetometer on your local intranet, displaying both the rolling 24 hour plots and yesterday's plots.

NOTE: if you have a different **hostname** for your RPi, change the search bar entry to…

http://*hostname*.local

Where ***hostname*** is the hostname for your RPi setup.

Code is also supplied that will enable the user to upload, via FTP, to their host website, should they have one - see **Optional remote FTP upload** section for more details.

Check PicoMagnetometerACM0 service is running

1. To check the **status** of your service, type the following command and press enter.

    sudo systemctl status PicoMagnetometerACM0.service

2. To **start** your service, type the following command and press enter.

    sudo systemctl start PicoMagnetometerACM0.service

3. To **stop** your service, type the following command and press enter.

    sudo systemctl stop PicoMagnetometerACM0.service

4. To **enable** your service, type the following command and press enter.

    sudo systemctl enable PicoMagnetometerACM0.service

5. To **disable** your service, type the following command and press enter.

    sudo systemctl disable PicoMagnetometerACM0.service

Optional Post install operational checks

After installing/updating scripts on your RPi, you can run:

sudo bash /home/pi/UKRAA_Magnetometer/scripts/runPostUpdateChecksACM0.sh

This executes:

- optional daily publish logic checks
- optional HDZ/BI web visibility checks

For daily runtime monitoring, a health marker is written by:

sudo bash /home/pi/UKRAA_Magnetometer/scripts/checkDailyPublishHealthACM0.sh

Marker output file:

/home/pi/UKRAA_Magnetometer/data/status/daily-health.txt

One-line dashboard summary on demand:

sudo bash /home/pi/UKRAA_Magnetometer/scripts/showDashboardSummaryACM0.sh

If scheduled in cron, summary snapshots can be collected in:

/home/pi/UKRAA_Magnetometer/logfiles/dashboard-summary.log

Optional Rolling alert Emails

Rolling alert evaluation supports user defined activity thresholds.

Default values will be:

- **Yellow** at **50nT**
- **Amber** at **100nT**
- **Red** at **200nT**

The alert evaluator runs as part of **processRollingData.sh** and only sends email on threshold transitions to levels you choose.

The same threshold values are also used by:

- Daily hourly Activity plot (**PlotDataActivityACM0.gp**)
- Rolling Activity plot (**PlotRollingActivityACM0.gp**)

Configure via .ini file (recommended)

1. Copy the example file: **install/alerts.ini.example**

2. Place it on the Pi as: **/home/pi/UKRAA_Magnetometer/config/alerts.ini**

3. Edit the values for your SMTP service and recipients.

4. Set activity thresholds in the same file under **[alerts]**:

    [alerts]
    yellow_threshold_nt = 50
    amber_threshold_nt = 100
    red_threshold_nt = 200

Enable or disable alert emails

Email sending is off by default so a fresh install does not log SMTP failures every 5 minutes.

In **alerts.ini**:

[alerts]
email_enabled = true

Notes:

- When **email_enabled = false**, thresholds, alert levels, plots and the web status page all still work; only email sending is skipped.
- Suppression is logged once per level transition, not on every run.
- The daily heartbeat email is also skipped while **email_enabled = false**.
- The manual test scripts (**testAlertEmailACM0.sh**, **testHeartbeatEmailACM0.sh**) still send, so you can verify SMTP before enabling.
- Environment variable override: **MAGNETOMETER_EMAIL_ENABLED**

By default, **EvaluateAlertsACM0.py** reads: **/home/pi/UKRAA_Magnetometer/config/alerts.ini**

You can override the config path with: **MAGNETOMETER_ALERTS_INI_PATH=/path/to/alerts.ini**

Configure which levels trigger email

Set **MAGNETOMETER_EMAIL_ALERT_LEVELS** as a comma-separated list:

- **RED**
- **RED,AMBER**
- **RED,AMBER,YELLOW**

Configure SMTP and recipients

You can set SMTP/email values in the **.ini** file, or by environment variables. Environment variables take precedence if both are set.

Available environment variables are:

- **MAGNETOMETER_SMTP_HOST** (required)
- **MAGNETOMETER_SMTP_PORT** (default 587)
- **MAGNETOMETER_SMTP_USERNAME** (optional)
- **MAGNETOMETER_SMTP_PASSWORD** (optional)
- **MAGNETOMETER_SMTP_STARTTLS** (***true*** by default)
- **MAGNETOMETER_SMTP_SSL** (***false*** by default)
- **MAGNETOMETER_EMAIL_FROM** (required)
- **MAGNETOMETER_EMAIL_TO** (required, comma-separated for multiple recipients)
- **MAGNETOMETER_EMAIL_ATTACH_PLOT** (***true*** by default)
- **MAGNETOMETER_WEB_URL** (optional, included in email body)

Notes

- Alert state is stored in **data/alerts/alert-state.json**
- Rolling status is read from **data/status/current.json**
- If **RollingActivity.png** exists, it is attached to the alert email
- Alert evaluator config precedence is: environment variable -> **.ini** file -> built-in default

Regenerate a chosen Activity date (quick local test mode)

You can regenerate the hourly Activity plot for any date without storing an archive copy by default:

/bin/bash /home/pi/UKRAA_Magnetometer/scripts/testActivityPlotACM0.sh YYYY-MM-DD

This updates only:

- **/home/pi/UKRAA_Magnetometer/temp/Activity.png**

To also write the dated archive file in **plots/Activity/**, add **--archive**:

/bin/bash /home/pi/UKRAA_Magnetometer/scripts/testActivityPlotACM0.sh YYYY-MM-DD --archive

Send a one-off SMTP test email

To verify SMTP settings without waiting for a threshold transition, run:

/bin/bash /home/pi/UKRAA_Magnetometer/scripts/testAlertEmailACM0.sh

Or run the Python script directly:

/usr/bin/python3 /home/pi/UKRAA_Magnetometer/scripts/EvaluateAlertsACM0.py --test-email

The test email does not update transition state, so normal alert logic is unaffected.

Optional daily heartbeat email

You can enable a once-per-day summary email so you know the system is alive even when no threshold transition occurs.

In **alerts.ini:**

[heartbeat]
enabled = true
hour_utc = 9
to =
attach_plot = false

Notes:

- **hour_utc** is the first UTC hour of the day when the heartbeat can send.
- Only one heartbeat attempt is made per UTC day (success or failure), to avoid retry spam every 5 minutes.
- If **heartbeat.to** is blank, it uses **[email] to**.

Environment variable overrides are also available:

- **MAGNETOMETER_HEARTBEAT_ENABLED**
- **MAGNETOMETER_HEARTBEAT_HOUR_UTC**
- **MAGNETOMETER_HEARTBEAT_TO**
- **MAGNETOMETER_HEARTBEAT_ATTACH_PLOT**

Send a one-off heartbeat test email immediately

To verify heartbeat email delivery without waiting for **heartbeat.hour_utc**, run:

/bin/bash /home/pi/UKRAA_Magnetometer/scripts/testHeartbeatEmailACM0.sh

Or run the Python script directly:

/usr/bin/python3 /home/pi/UKRAA_Magnetometer/scripts/EvaluateAlertsACM0.py --test-heartbeat

This immediate heartbeat test does not update daily heartbeat schedule/state tracking.

Optional Remote FTP upload

You can upload plot PNG files to an external hosting site while keeping local web publishing as the primary path.

Remote upload config file:

- **/home/pi/UKRAA_Magnetometer/config/remote-upload.ini**

Template created by installer:

- **install/remote-upload.ini.example**

Example settings:

```
[ftp]
enabled = true
site = your-uploader-fp
user = your-upload-user
password = your-upload-password
port = 21
directory = /data
timeout_seconds = 30
passive = true
create_dirs = true
upload_status_json = false
```

Behaviour:

- Daily run (09:30 via moveGraphs.sh) uploads:
    - **Activity.png**, **XYZ.png**, **HDZ.png**, **BI.png** to **/data**
- Rolling run (every 5 minutes via processRollingData.sh) uploads:
    - **RollingActivity.png**, **RollingXYZ.png, RollingHDZ.png**, **RollingBI.png** to **/data/rolling**
- Optional rolling status JSON upload:
    - set **upload_status_json = true**
    - uploads **data/status/current.json** to **/data/status/current.json**
    - external webpage status fallback (**status/current.json**) requires this to be true.
- Remote upload failures are non-blocking and do not stop local publishing.

Manual test commands:

```
/bin/bash /home/pi/UKRAA_Magnetometer/scripts/uploadRemoteACM0.sh daily
/bin/bash /home/pi/UKRAA_Magnetometer/scripts/uploadRemoteACM0.sh rolling
```

Combined test helper:

```
/bin/bash /home/pi/UKRAA_Magnetometer/scripts/testRemoteUploadACM0.sh
```


License

Contact us

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UK Radio Astronomy Association